

Stephen Remi Adebambo

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SKILLS

- **Programming:** Python, TypeScript, JavaScript, C, SQL
- **Technologies:** React, Next.js, Node.js, FastAPI, PostgreSQL, Redis, Docker, Terraform, Airflow, Snowflake, LangGraph, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Tailwind CSS, Linux, Nix/NixOS
- **Professional:** Full-Stack Development, Machine Learning, Data Engineering, System Architecture, API Design-, Predictive Modeling, Reinforcement Learning, Agile/Scrum, Infrastructure Automation, CI/CD, Data Visualization, Feature Engineering

EXPERIENCE

Prof. Mark Crowley's Lab

Research Assistant

Jun 2026 - Present

Waterloo, ON, Canada

- Adapted individual reinforcement learning models into multi-agent training methods to evaluate scalability of normative safety findings in simulated resource-competition environments.
- Tested fragility of parameter choices by simulating how different world specifications affect agent behavior and prevent ethical drift.
- Investigated instability of static reward signals and peer feedback to mitigate hallucination risks as agents operate in changing environments.

Lanstev Fashion

Bookkeeping Assistant

Mar 2026 - Jun 2026

Remote

- Eliminated manual entry errors by developing an automated spreadsheet system to reconcile costs, profit, and expense metrics.
- Recovered missing financial records by auditing historical transactions and structuring them into a centralized database.
- Reduced monthly administrative overhead by implementing formula-driven expense tracking workflows.

The Electoral Hub

Data Analyst / IT Officer

Feb 2025 - Mar 2025

Abuja, Nigeria

- Accelerated political report generation from multiple days to under four hours by automating regional data processing pipelines in Python.
- Clarified electoral trends for non-technical audiences by building Python data pipelines to generate regional visualizations.
- Reduced report generation time by over 80% by replacing manual processing with semi-automated Python workflows.

National Productivity Centre

Office Clerk (NYSC)

Jan 2024 - Oct 2024

Abuja, Nigeria

- Supported daily departmental operations, measured by uninterrupted service delivery, by managing communications between students and coworkers.
- Formatted official documents, measured by error-free record keeping, by typing and reviewing regular correspondences.

Multichoice Nigeria

Network Operations Intern

Jul 2022 - Sep 2022

Abuja, Nigeria

- Monitored server deployments and network infrastructure using Linux command line tools, maintaining near-constant uptime for internal business units.
- Prevented service disruptions by identifying and independently resolving 11 connectivity issues.
- Diagnosed routing problems with Traceroute and routing tables while collaborating with senior technicians, restoring network service within 30 minutes on average and reducing escalation time.

Aiivon Innovation Hub

Software Developer Intern

Jul 2021 - Sep 2021

Abuja, Nigeria

- Built fintech applications and internal monitoring tools using TypeScript and React, collaborating with product designers to create responsive Tailwind CSS front-ends and RESTful APIs, which enabled the team to launch new features on schedule.
- Increased user retention by 10% by resolving component rendering bugs and overhauling user interfaces.
- Decreased codebase size by 6% by refactoring monolithic code into modular patterns to improve maintainability.

PROJECTS

World Cup 2026 Performance Dashboard [GitHub](#)

May 2026 - Present

- Processed live sports data by building a Python ETL pipeline using Celery and Airflow to hydrate the PostgreSQL database.
- Calculated pre-match win probabilities by integrating a machine learning prediction engine trained on historical statistics.
- Stored live match events by designing a 3NF PostgreSQL schema with read-optimized materialised views.
- Achieved real-time frontend updates by implementing Server-Sent Events in a Next.js App Router environment.
- Delivered scrolling performance across large datasets by integrating TanStack Virtual headless UI tables.

InfraLens [GitHub](#)

Apr 2026

- Identified cheaper cloud droplet sizes by applying heuristic analysis on p95 CPU and memory utilization within Snowflake.
- Automated cloud infrastructure rightsizing by building a FastAPI and Snowflake pipeline to analyze DigitalOcean CPU and memory utilization metrics.
- Explained infrastructure recommendations by integrating Llama-3.1-70b and Nemotron-12b language models into the data pipeline.
- Generated automatic Terraform pull requests by integrating LLM explanation agents with a Next.js websocket frontend to justify infrastructure changes.
- Award: MLH Best Use of DigitalOcean Award, ConHacks 2026.

CityMind [GitHub](#)

Mar 2026

- Enforced role-based access control on municipal datasets by developing an API gateway using React and FastAPI.
- Maintained query auditability and cell-level suppression by routing natural language requests through a LangGraph AI agent against a read-only SQLite replica.
- Award: 1st Place, City Data OS Problem Statement, FCI x LangChain Hackathon.

MazeEnv [GitHub](#)

Mar 2026 - Apr 2026

- Benchmarked temporal difference control algorithms by developing a 10x10 grid-world maze environment using Python and NumPy to analyze convergence and variance.
- Identified off-policy recklessness in Q-Learning by evaluating algorithms across hazard-dense configurations over 1500 episodes.
- Quantified reinforcement learning suboptimality by demonstrating that generated paths were 33-63% longer than classical A* search solutions.

EDUCATION

University of Waterloo
MEng., Electrical and Computer Engineering
• **GPA:** 87.33%
• **Coursework:** Algorithms and Complexity Analysis, Artificial Intelligence, Machine Learning

Sep 2025 - Apr 2027
Waterloo, ON, Canada

Nile University of Nigeria
BSc, Computer Science
• **GPA:** 4.76/5.0
• **Achievements:** Graduated with First Class Honours, Full-Scholarship Recipient

Sep 2019 - Sep 2023
Abuja, Nigeria